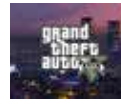




JioBook in India

The new JioBook laptop with Windows 10 may be closer to its launch date in India. <https://digit.in/mar22-51>



GTA5 on Playstation and Xbox

GTA 5 is set to arrive on the next-gen PlayStation 5 and Xbox Series S/X consoles on March 15. <https://digit.in/mar22-52>

The Efficiency of Spiderwebs

A spider uses mathematics to optimise its meal-making process.

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If you see a spider in the corner of your room, your first instinct might be to throw something at it.

And we don't blame you, about 3 to 15 per cent of the population has arachnophobia, the fear of spiders. The perception of spiders as astute, useful insects is not prevalent, with most people developing a disgust towards them. This disgust and fear stems from the "creepy-crawly" association that people make with a spider's movement. However, this creepy-crawly movement proves to be one of the most mysterious algorithms in terms of mathematical computation and simulation. Building a spiderweb is a great feat of engineering that many scientists have unveiled the secrets of only recently.

From a general point-of-view, a simple understanding of the spider-web exists in every individual's mind. But there are so many aspects to building, sustaining, and evolving a spider-web. The basic function of a spider-web is to capture their prey for feeding. Spiders release threads that are made of a unique silk-like substance commonly referred to as "spider silk".

SPIDER SILK: WHAT WE KNOW

Spider silk is a glossy, strong, and light

substance with chemical properties. It is tougher than steel and has a high tensile strength, which means it can withstand a lot of stretching before snapping. An individual spider may have more than five separate silk glands, depending on the species, where silks of various qualities are created before being extruded by three pairs of spinnerets. The dragline is extruded from the ampullate gland through the front spinnerets. Spider dragline is the strongest and toughest of a spider's silks, and it serves as the web's structural framework.

Orb-weaving spiders have unique glands that may produce up to seven different types of silk. Thin or thick silk threads, dry or sticky, beaded or smooth, are all possibilities. The strands that a spider uses to weave its web start out as liquid, but they quickly dry out in the air. Some spiders have the ability to manufacture multiple types of silk. A typical orb-web, for example, may have at least four types, each of which adds a different feature, such as strength, flexibility, or stickiness. Scientists have spent decades

attempting to figure out what gives silk its strength and flexibility, but they have only uncovered hints so far.

HOW DOES THE SPIDER DO IT?

When a spider spins a web, the silk thread is released. Wherever it constructs its web, it hooks the thread to something – a branch, a room corner, a doorframe. The spider adds new threads to the web as it goes back and forth, strengthening it and forming a pattern. "Radial lines" are lines that radiate outward from the web's centre. They help to uphold the web's multiple layers. "Orb lines" are web threads that loop around and around.

A STRUCTURAL MARVEL

Spider webs are a natural example of tensegrity (tensional integrity) structures, a type of prestressed, wired system. Due to the perfect distribution of structural mass, these webs represent a unique marriage of geometry and mechanics, resulting in extremely efficient structures. The geometry of a tensegrity structure is crucial in determining its existence and stiffness at a